Code: 20A04602T

R20

B.Tech III Year II Semester (R20) Supplementary Examinations November/December 2024

VLSI DESIGN

(Electronics and Communication Engineering)

Time: 3 hours   Max. Marks: 70

PART – A

(Compulsory Question)

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1		Answer the following: (10 X 02 = 20 Marks)	
	(a)	List out few comparisons between CMOS and BiCMOS transistors.	2M
	(b)	Define g_m of MOS transistor.	2M
	(c)	List out the different stages in VLSI Design flow.	2M
	(d)	Mention the differences between stick diagram and layout diagram.	2M
	(e)	What are the issues involved in driving large capacitive loads in VLSI circuits?	2M
	(f)	Draw the Transmission gate symbol and write its properties.	2M
	(g)	Draw 2-bit comparator.	2M
	(h)	On what basis FPGA's are classified?	2M
	(i)	Define controllability and observability with respect to testing.	2M
	(j)	Mention the principle of Built in self-test.	2M

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

2		Draw the fabrication steps of NMOS transistor and explain its operation in detail.	10M
		OR	
3		Derive the Drain to source current I_{ds} versus voltage V_{ds} relationships in Non saturated and saturated regions.	10M
4	(a)	Draw the Stick diagram and layout diagram for NAND gate using CMOS.	5M
	(b)	Discuss the limits due to sub-threshold currents and current density.	5M
		OR	
5	(a)	Explain Lambda based Design rules for contact cuts with neat diagrams.	5M
	(b)	What are the general observations on the design rules?	5M
6		What are the Alternate gate circuits available? Explain any one of them with suitable sketch by taking NAND gate as example.	10M
		OR	
7	(a)	Derive the rise-time, fall-time of CMOS inverter Delay with neat diagrams.	5M
	(b)	Explain Sheet resistance, Standard unit of capacitance c_g with suitable examples.	5M
8	(a)	Draw the FPGA architecture and explain the function of each block.	5M
	(b)	Describe the design procedure of an asynchronous counter.	5M
		OR	
9	(a)	Explain the influence of different Peremeters on low power design.	5M
	(b)	Explain the operation of Booth multiplication with suitable example.	5M
10	(a)	Explain sensitized path-based testing with an example.	5M
	(b)	Explain the effect of memory in testing sequential logic.	5M
		OR	
11		Explain in detail about design for testability.	10M

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